

Topic 8 - Individual Demand

Part II:

i) Income & Substitution Effects

ii) Decomposition of a price change into Income & Substitution Effects

iii) Analytical Calculation of IE and SE

ECON 101
Spring 22

Suppose the good in question is amount of rent paid / apartment vs other goods

- Suppose there is a decrease in the rental price of your unit.

↳ rent involves a large share of your income.

↳ you will end up feeling richer because you will have more disposable income to buy other stuff

↳ Example of an Income effect. (Pure)

What fruit to buy? Grunny Smith apples (green, tart taste)
↳ this involves very little share of your income. OR Red Delicious apples (bright red, little taste)

If you are equally happy with any apple → you end up buying

the one that is cheaper → if the prices change you buy the one with the lowest price, i.e. substituting towards the relatively cheaper.

Pure Substitution

Calculating Income & Substitution Effects: Graphical Description.

- Suppose we want to analyze the change in quantity demanded of good 1 after a price change, from P_1^0 to P_1^1 , while holding the price of good 2 and income fixed (say at $\bar{P}_2$ and $\bar{I}$).

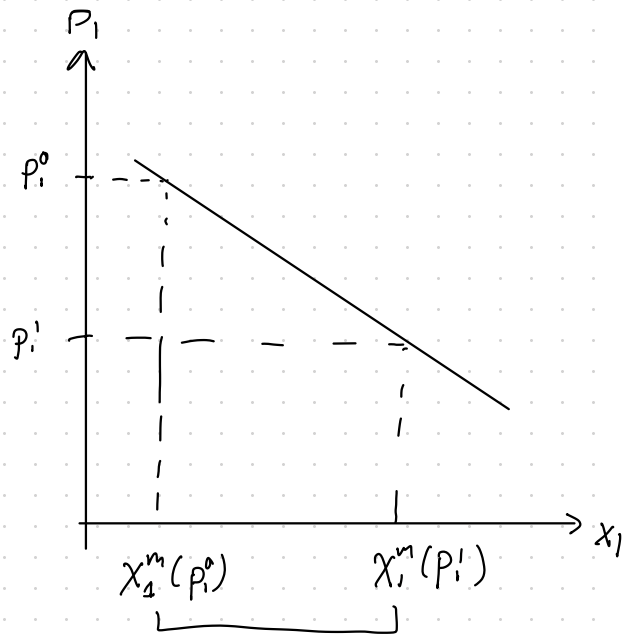
The total effect is: $TE = X_1(P_1^1, \bar{P}_2, \bar{I}) - X_1(P_1^0, \bar{P}_2, \bar{I})$

or for simplicity: $TE = X_1(P_1^1) - X_1(P_1^0)$ [Change in X_1 across its individual demand curve]

→ The goal is to decompose the TE into the Substitution Effect (SE) and the Income Effect (IE)

→ The SE respects the law of demand (goes in opposite direction to the price change)

→ The IE depends on whether the good is normal or inferior.



$$TE = x_1^m(P^1) - x_1^m(P^0)$$

$$TE = SE + IE$$

Consider a price decrease $\Delta P_1 < 0$

- $SE > 0$ (Satisfies the law of demand)

Opposite sign always

to ΔP_1

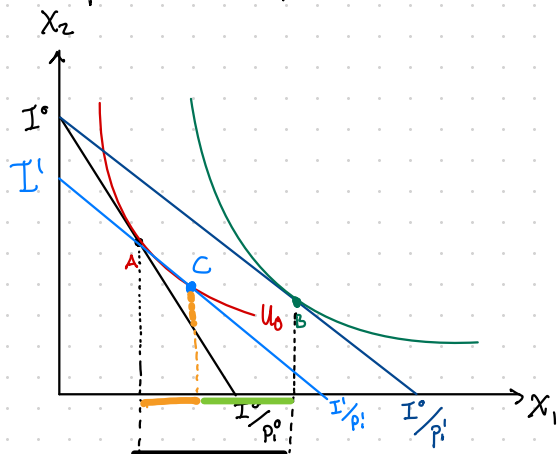
- $IE \begin{matrix} > 0 \\ = 0 \\ < 0 \end{matrix}$

$$TE = \begin{cases} > 0 & \text{if } IE \text{ in same direction as } SE \\ & \text{or } SE \text{ bigger than } IE \\ < 0 & \text{if } SE \text{ opposite } IE \text{ and } IE \text{ bigger.} \end{cases}$$

Case 1A: Normal good. and a price decrease.

$$p_1^0 > p_1' \Rightarrow \Delta p_1 = p_1' - p_1^0 < 0; \quad \bar{p}_2 = 1, \quad I^0 \text{ (original level of income)}$$

$$-TE = x_1^m(p_1', 1, I^0) - x_1^m(p_1^0, 1, I^0)$$



$$A = (x_1^m(p_1^0, 1, I^0), x_2^m(p_1^0, 1, I^0))$$

$$B = (x_1^m(p_1', 1, I^1), x_2^m(p_1', 1, I^1))$$

$$U^0 = U(A) = U(x_1^m(p_1^0, 1, I^0), x_2^m(p_1^0, 1, I^0))$$

$$\text{Solve } \min_{x_1, x_2} p_1' x_1 + x_2(1) \quad \text{s.t.} \quad U(x_1, x_2) = U^0 \quad \Rightarrow C \text{ is the solution to this problem.}$$

Calculate the cost bundle C at the new prices: $p_1' x_1^C + (1) x_2^C = I^1$

C can also be computed as: $x_1^m(p_1', 1, I')$
 $x_2^m(p_1', 1, I')$

Income Compensation $\Delta I = I' - I^0 < 0$

$$-SE: x_1^m(p_1', 1, I^1) - x_1^m(p_1^0, 1, I^0) > 0$$

$$-IE: x_1^m(p_1', 1, I^0) - x_1^m(p_1', 1, I^1) > 0$$

$$TE = SE + IE$$

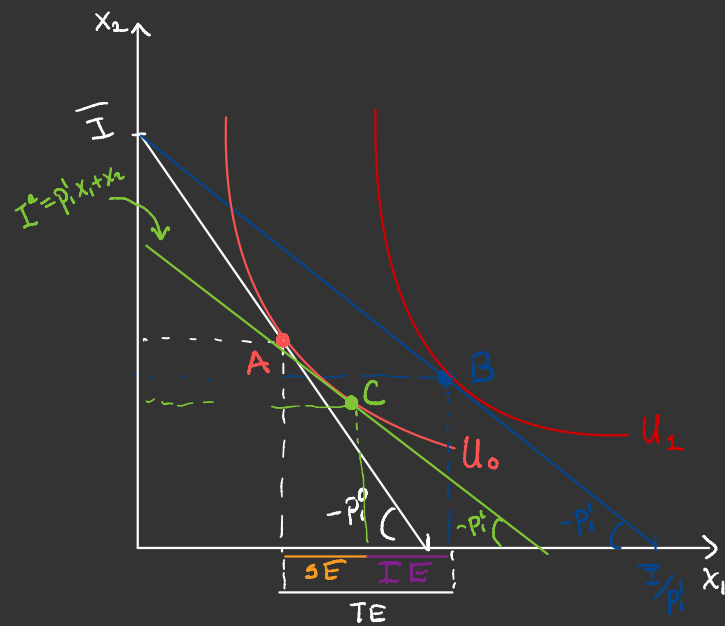
For a normal good the TE effect satisfies the law of demand.

Remark: In SE, U^0 is held fixed, as well as p_2 only p_1 is changing!

→ In order to hold utility fixed, income should be compensated.

Case 1A: Normal good and a price decrease.

$$p_1^0 > p_1^1 \Rightarrow \Delta p_1 = p_1^1 - p_1^0 < 0 ; p_2 = 1, I = \bar{I}$$



$$A = (x_1(p_1^0, 1, \bar{I}), x_2(p_1^0, 1, \bar{I}))$$

$$B = (x_1(p_1^1, 1, \bar{I}), x_2(p_1^1, 1, \bar{I}))$$

$$C = (x_1(p_1^1, 1, I^a), x_2(p_1^1, 1, I^a))$$

$$TE = x_1(p_1^1, 1, \bar{I}) - x_1(p_1^0, 1, \bar{I}) > 0$$

To decompose the TE, since $\Delta p_1 < 0$ (price decrease) we need to take money away from the consumer to leave her at utility level U_0 : the green budget line has income $I^a < \bar{I}$. Thus we take money away from the consumer $\bar{I} - I^a = \Delta I$. ΔI is the compensation: $\bar{I} - \Delta I = I^a$

$$SE = x_1(p_1^1, 1, I^a) - x_1(p_1^0, 1, \bar{I}) > 0$$

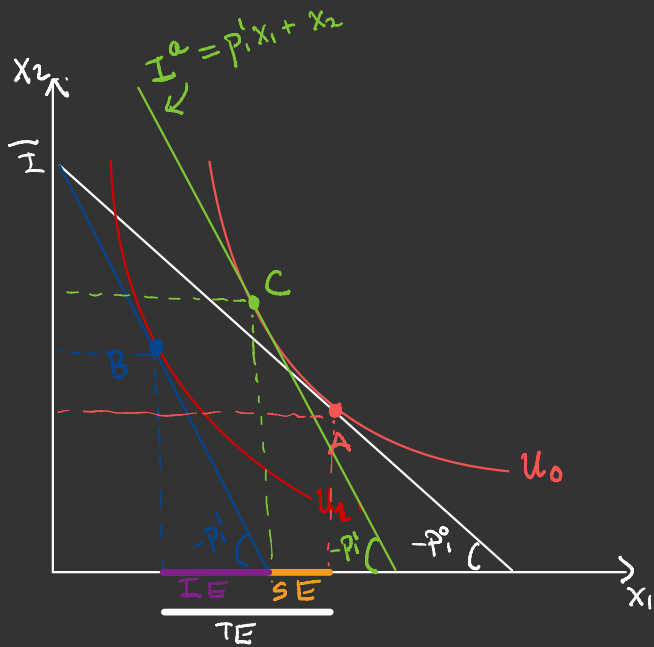
$$IE = x_1(p_1^1, 1, \bar{I}) - x_1(p_1^1, 1, I^a) > 0$$

$$TE = x_1(p_1^1, 1, \bar{I}) - x_1(p_1^0, 1, \bar{I}) > 0$$

For a normal good the TE will always satisfy the law of demand because the SE and IE have the same sign!

Case 1B: Normal good and a increase in price p_1

$$p_1^0 < p_1^1 \Rightarrow \Delta p_1 = p_1^1 - p_1^0 > 0 ; p_2 = 1, I = \bar{I}$$



$$A = (x_1(p_1^0, 1, \bar{I}), x_2(p_1^0, 1, \bar{I}))$$

$$B = (x_1(p_1^1, 1, \bar{I}), x_2(p_1^1, 1, \bar{I}))$$

$$C = (x_1(p_1^1, 1, I^a), x_2(p_1^1, 1, I^a))$$

$$TE = x_1(p_1^1, 1, \bar{I}) - x_1(p_1^0, 1, \bar{I}) < 0$$

To decompose the TE, since $\Delta p_1 > 0$ (price increase) we need compensate the consumer by giving her more money to leave her at utility level U_0 : the green budget line has income $I^a > \bar{I}$, thus we give extra income to the consumer $I^a - \bar{I} = \Delta I$. ΔI is the compensation: $\bar{I} + \Delta I = I^a$

$$SE = x_1(p_1^1, 1, I^a) - x_1(p_1^0, 1, \bar{I}) < 0$$

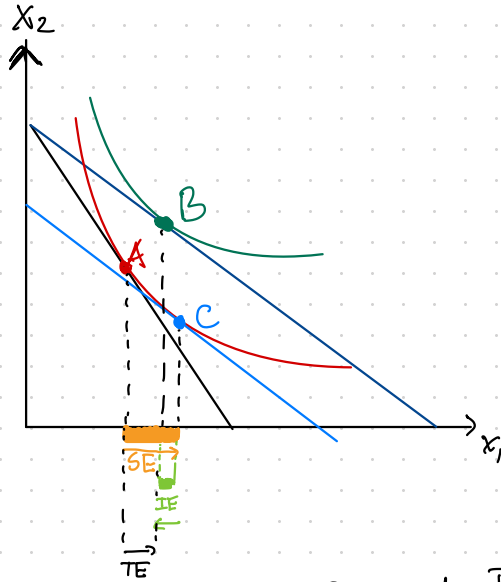
$$IE = x_1(p_1^1, 1, \bar{I}) - x_1(p_1^1, 1, I^a) < 0$$

$$TE = x_1(p_1^1, 1, \bar{I}) - x_1(p_1^0, 1, \bar{I}) < 0$$

Again since x_1 is normal, SE and the IE have the same sign.

Case 2: Inferior good. (price decrease)

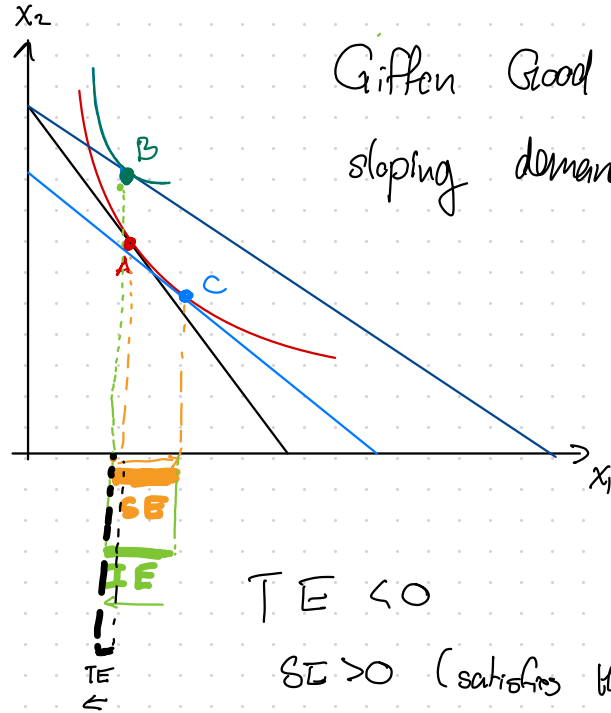
$$p_1^0 < p_1^1 \Rightarrow \Delta p_1 < 0, \quad \bar{p}_2 = 1, \quad I = I^0$$



$$TE > 0$$

BUT $IE < 0$
 $SE > 0$

SE and IE
 go in opposite
 directions
 but SE dominates
 IE



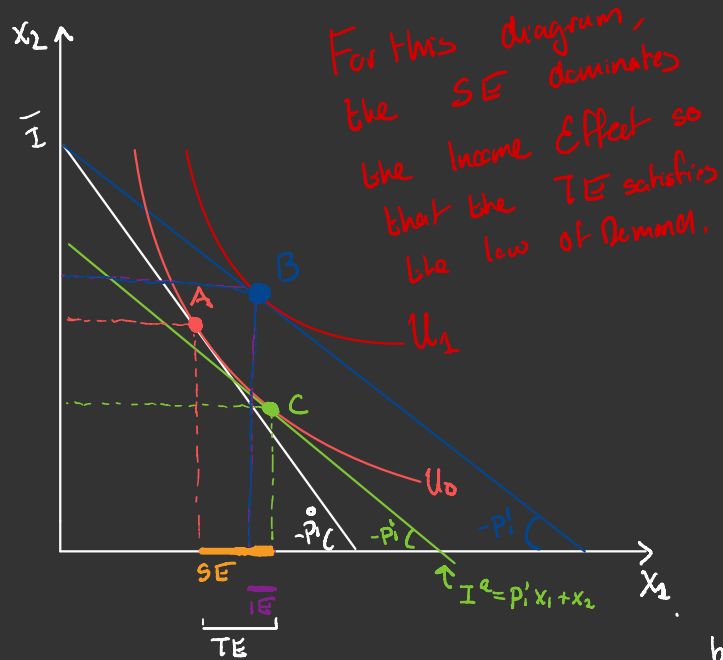
Giffen Good (upward
 sloping demand)

$$TE < 0$$

$SE > 0$ (satisfies the Law of demand)
 $IE < 0$ and dominates the SE
 $\Rightarrow TE < 0$

Case 2 : Inferior good (I only analyze a price decrease)

$$p_1^0 > p_1^1 \Rightarrow \Delta p_1 = p_1^1 - p_1^0 < 0 ; p_2 = 1, I = \bar{I}$$



Since this is a price decrease ($\Delta p_1 < 0$) again we need to take money away

from the consumer:

$\bar{I} - I^e = \Delta I$; ΔI is the amount we need to take away; $\bar{I} - \Delta I = I^e$. I^e is the auxiliary income

$$SE = x_1(p_1^1, 1, I^e) - x_1(p_1^0, 1, \bar{I}) > 0$$

$$IE = x_1(p_1^1, 1, \bar{I}) - x_1(p_1^1, 1, I^e) < 0$$

because the good is inferior

$$A = (x_1(p_1^0, 1, \bar{I}), x_2(p_1^0, 1, \bar{I}))$$

$$B = (x_1(p_1^1, 1, \bar{I}), x_2(p_1^1, 1, \bar{I}))$$

$$C = (x_1(p_1^1, 1, I^e), x_2(p_1^1, 1, I^e))$$

$$TE = x_1(p_1^1, 1, \bar{I}) - x_1(p_1^0, 1, \bar{I}) = ?$$

If $|SE| > |IE|$ then the total effect is positive, as expected from a price decrease

Thus, for an inferior good, SE and IE have opposite signs!

Example: Income and Substitution Effects from the Utility function
 $U(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$, $0 < \alpha < 1$.

Recall that the MRS for this utility function is $MRS = \frac{\alpha}{1-\alpha} \frac{x_2}{x_1}$
 Recall also that the demand system that is obtained from solving the utility maximization problem

$$\max_{(x_1, x_2)} x_1^\alpha x_2^{1-\alpha}$$

subject to $p_1 x_1 + p_2 x_2 = I$

is: $x_1(p_1, p_2, I) = \frac{\alpha I}{p_1}$; $x_2(p_1, p_2, I) = \frac{(1-\alpha) I}{p_2}$

Suppose that $p_2 = 1$, $\bar{I} = 100$. We will consider a price decrease from $p_1^0 = 2$ to $p_1^1 = 1$, $\Delta p_1 = p_1^1 - p_1^0 = -1 < 0$. Suppose also that $\alpha = 0.3$

$$x_1(2, 1, 100) = \frac{0.3 \times 100}{2} = 15; \quad x_2(2, 1, 100) = \frac{0.7 \times 100}{1} = 70$$

$$x_1(1, 1, 100) = \frac{0.3 \times 100}{1} = 30; \quad x_2(1, 1, 100) = \frac{0.7 \times 100}{1} = 70$$

$$U_0 = U(A) = 15^{0.3} \times 70^{0.7} \approx 44.0957; \quad U_0 = x_1^{0.3} x_2^{0.7}$$

$$U_1 = U(B) = 30^{0.3} \times 70^{0.7} \approx 54.2881; \quad U_1 = x_1^{0.3} x_2^{0.7}$$

To find I^a , we use the expenditure minimization problem:

$$\min_{(x_1, x_2)} (1)x_1 + (1)x_2 \quad \text{these are expenditures at the new prices } p_1=1, p_2=1$$

subject to $x_1^{0.3} x_2^{0.7} = U_0$ } the optimality condition is the same: $MRS = p_1/p_2$

$$\frac{0.3}{0.7} \frac{x_2}{x_1} = \frac{1}{1} \Rightarrow \frac{3}{7} \frac{x_2}{x_1} = 1 \Rightarrow x_2 = \frac{7}{3} x_1$$

Substitute $x_2 = \frac{7}{3} x_1$ into the utility constraint:

$$x_1^{0.3} \left(\frac{7}{3} x_1\right)^{0.7} = U_0 \Rightarrow x_1^{0.3+0.7} \left(\frac{7}{3}\right)^{0.7} = U_0 \Rightarrow$$

$$x_1 = \frac{U_0}{\left(\frac{7}{3}\right)^{0.7}} \approx 24.3676, \quad x_2 = \frac{7}{3} (24.3676) \approx 56.8577$$

$$I^a: \text{minimal level of income} \Rightarrow I^a = (1)(24.3676) + (1)(56.8577) = 81.2253$$

required to achieve U_0

$$\bar{I} - I^a = \Delta I = 100 - 81.2253 = 18.7747$$

$$\text{Note: } x_1(1, 1, 81.2253) = \frac{0.3 \times 81.2253}{1} \approx 24.3676$$

$$x_2(1, 1, 81.2253) = \frac{0.7 \times 81.2253}{1} \approx 56.8577$$

$$SE = 24.3676 - 15 = 9.3676$$

$$IE = 30 - 24.3676 = 5.6324$$

$$TE = 30 - 15 = 9.3676 + 5.6324$$

